CANDIDATE NAME: ________________________________ INDEX NUMBER _________

CG _________

SERANGOON JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION 2018

H2 BIOLOGY **9744/02**
Paper 2 Structured Questions

Thursday
13 September 2018

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name, index number and CG in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [ ] at the end of each question or part question.

FOR EXAMINER'S USE	
Paper 1 (MCQ)	**/30**
Paper 2	
1	**/18**
2	**/17**
3	**/11**
4	**/13**
5	**/12**
6	**/9**
7	**/8**
8	**/12**
P2 Total	**/100**
Paper 3	**/75**
Paper 4	**/55**
TOTAL (100%)	

This question paper consists of 18 printed pages including this cover page.

1. Fig. 1.1 is an electron micrograph of part of an animal cell obtained from a rodent.

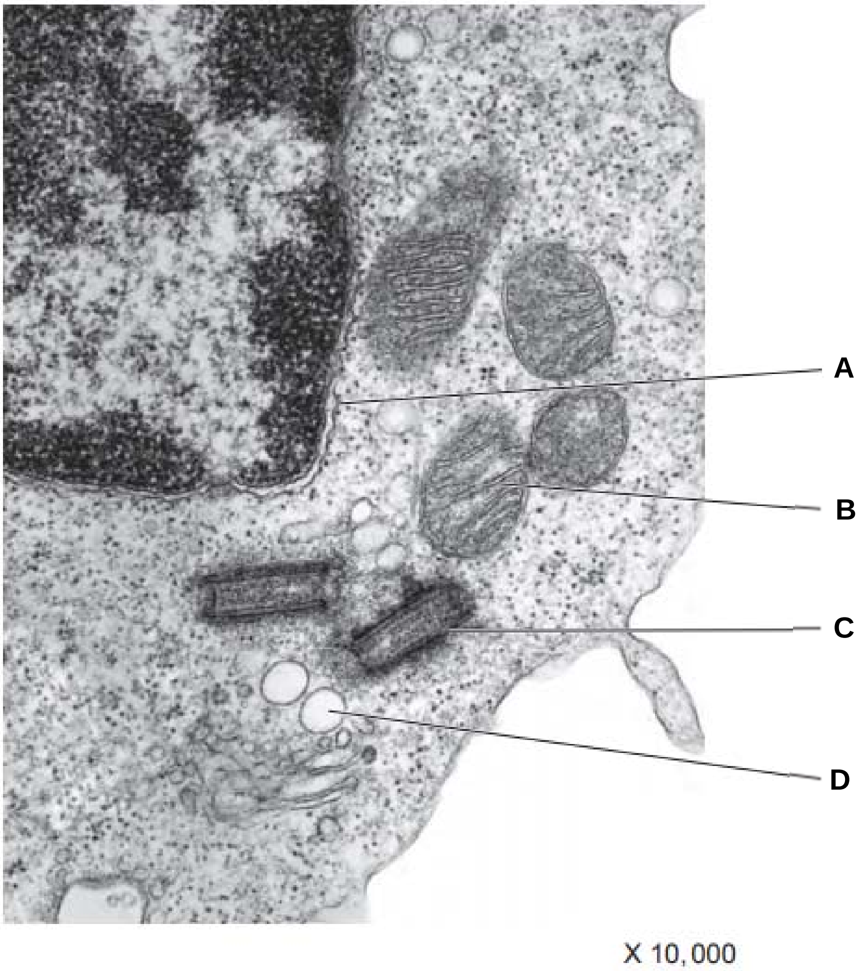


Fig. 1.1

(a) Name the structures labelled **A** to **C**. [3]

A: nuclear envelope/membrane / outer nuclear envelope/membrane / nucleus

B: crista/ mitochondrion/ inner mitochondrion membrane Rej: Matrix (image is clear enough)

C: centriole

(b) Describe the role of structure **C** in animal cells. [2]

1. Assembly/ organisation of <u>spindle fibres/ microtubule</u>
2. To separate <u>chromosomes/ chromatids</u> during <u>mitosis and meiosis/ cell division</u>
3. AVP: Modified centrioles can be found in <u>flagella/ cilia</u> for locomotion

(c) Complete karyotypes of two rodents of the same species were isolated as shown in Fig. 1.2. In this species of rodent, males are the heterogametic sex, where they have two different sex chromosomes.

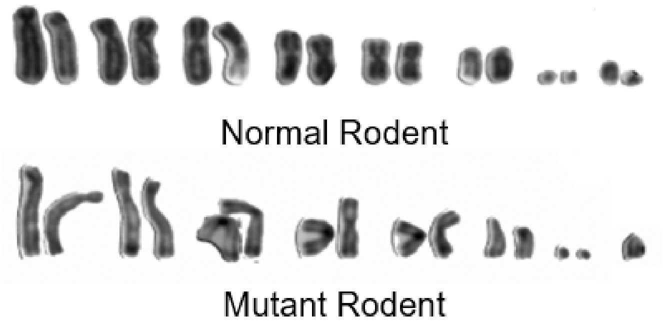


Fig. 1.2

(i) State the diploid number of chromosomes in a normal rodent. [1]

16

(ii) Explain how the mutant rodent karyotype was formed. [3]

1. Mutant rodent is missing one <u>sex chromosome</u> (i.e. XO)
2. <u>Nondisjunction</u> occurred during <u>anaphase</u> of meiosis I for the sex chromosome pair during formation of gametes in mutant rodent's father/ mother
3. Both sex chromosomes were distributed to one daughter cell at the end of <u>meiosis I</u>, while none were distributed to the other, forming <u>mutant gametes</u>
4. Fusion of mutant gamete (n-1) with a normal gamete during fertilisation led to formation of mutant rodent karyotype.

Max 3

(iii) Suggest why the mutant rodent is still able to survive despite the mutation. [1]

1. Only one copy of the X chromosome is required to express genes on the X chromosome sufficiently

Fig. 1.3 is an electron micrograph of a cancer cell in the midst of mitosis.

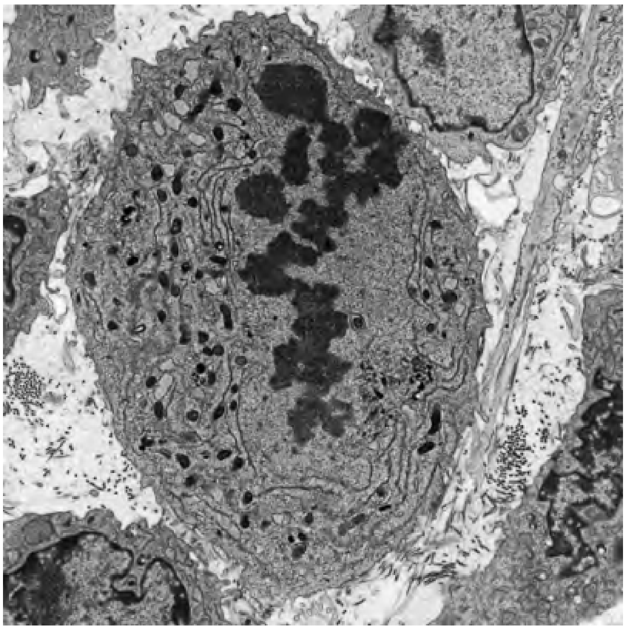


Fig. 1.3

(d) Identify the stage of mitosis, giving a reason for your answer. [2]

1. Stage: Metaphase
2. Reason: chromosomes align in a single/one row at the equator

People who have smoked cigarettes for many years are at risk of developing lung cancer.

(e) Explain why smoking increases the risk of cancer. [2]

1. Cigarettes contain tar/ nitrosamines/ AVP which is a carcinogen
2. That increases rate of mutation in tumor suppressor genes/ proto-oncogenes

Fig. 1.4 shows the change in the percentage of smokers in the male population of the UK between 1950 and 2005. Fig. 1.5 shows the change in mortality rate in the UK in men aged 75 to 84 between 1950 and 2005.

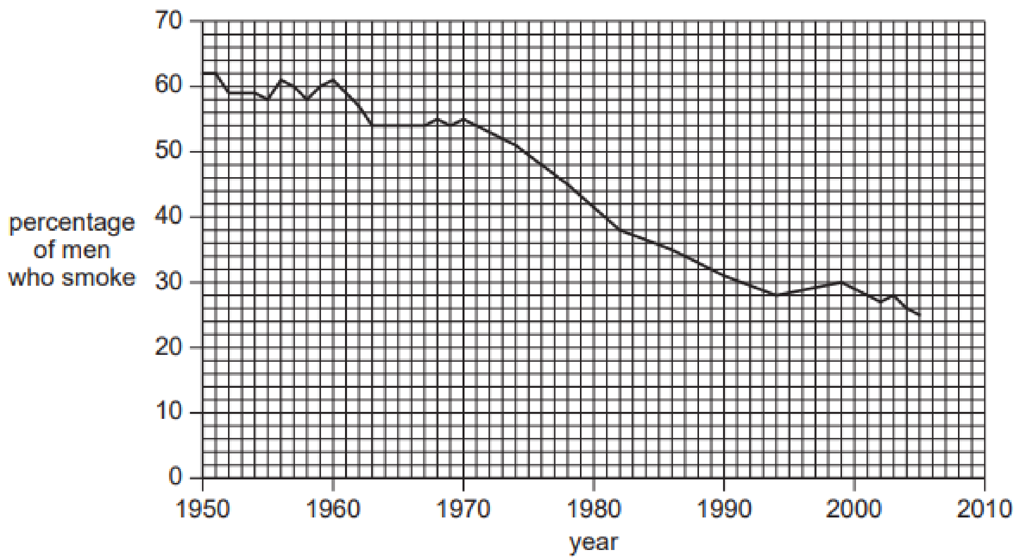


Fig. 1.4

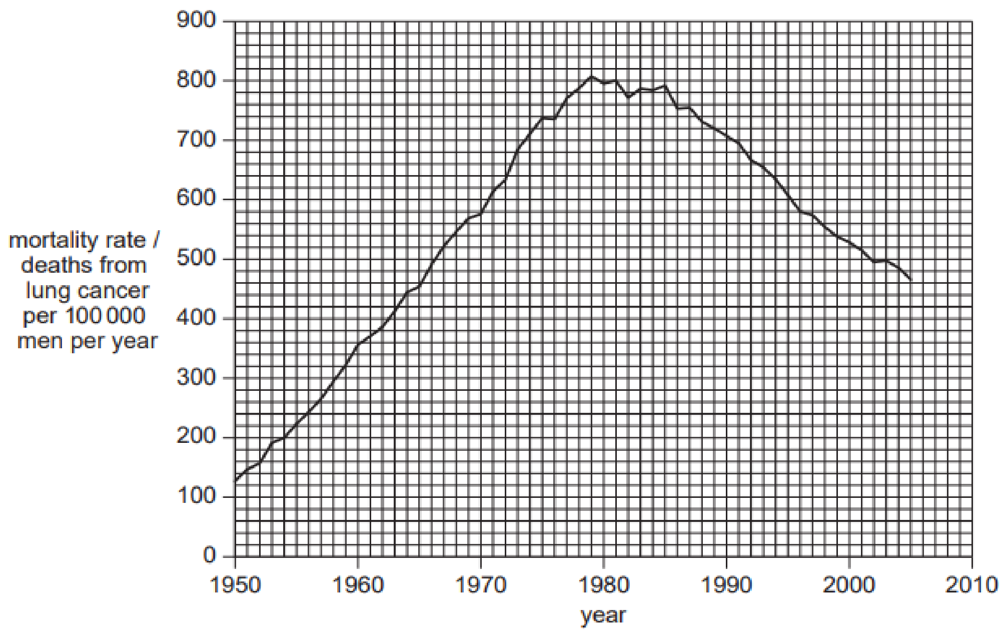


Fig. 1.5

(f) With reference to Figs. 1.4 and 1.5, discuss the observations made between 1950 and 2005. [4]

1. Percentage of men who smoke fell from 62 to 25% between 1950 and 2005
2. and mortality rate increased from 130 to 810 deaths from lung cancer per 100 000 men per year between 1950 and 1979 and then decreased to 470 in 2005
3. Increase in number of deaths despite fall in % smokers between 1950 and 1979 as cancer is multistep process/ requires multiple mutations to accumulate before it develops into cancer hence lag time/ death from cancer is often associated with metastasis which occurs typically in the later stages of cancer development
4. fall in cancer deaths due to improvements in health care that allow for earlier diagnosis/ treatment/AVP

[Total: 18]

2. **(a)** Contrast between the structures of DNA and tRNA. [3]

	DNA	tRNA
Number of strands	Double stranded/ two strands	Single stranded
structure	helix	Clover-leaf
monomer	deoxyribonucleotide	ribonucleotide
bases	Thymine present	Thymine absent, uracil is used instead
H bond frequency	H bond occurs between all bases	H bond only at selected regions to hold structure
A:T/U G:C ratio	A:T and G:C ratio is always 1	A:T/U and G:C ratio varies
length	longer	shorter

(b) Table 2.1 shows two mRNA triplets. Fill in the complementary tRNA triplets in the spaces provided. [1]

mRNA triplets	CGC	AAC
tRNA triplets	GCG	UUG

Table 2.1

(c) Calculate the minimum number of DNA nucleotides required to code for a polypeptide with 238 amino acids. Show your working. [2]

1. 238 x 3 = 714
2. Stop codon: 3 bases, Total: 714 + 3 = 717

(d) Describe the role played by tRNA in polypeptide synthesis. [3]

1. Carries amino acid to tRNA binding sites on ribosome
2. Bring amino acids in close proximity for formation of peptide bond
3. Specific amino acid sequence ensured by complementary base pairing of anticodon on tRNA with codon on mRNA

Eukaryotic cells are able to regulate gene expression to ensure that resources within the cell are used effectively and efficiently.

(e) Regulation can be observed during the process of stem cell specialisation. Describe the most likely form of regulation taking place during stem cell specialisation, giving reasons for your answer. [4]

1. DNA methylation at cytosine residues in CpG island in promoter catalysed by DNA methyltransferases
2. Results in increased condensation of chromatin and is associated with long term and heritable inactivation of genes
3. During specialisation, stem cells differentiate to form specialised cells that perform specific functions
4. Differentiated cells no longer need to express a wide repertoire of genes involved in maintaining stem cell properties hence can be permanently inactivated
5. Modification must be heritable to allow differentiated cell to produce identical differentiated daughter cells by mitosis

Max 4

(f) Regulation can also occur following the synthesis of proteins. An example of such regulation is shown in Fig. 2.1 where inactive enzyme precursor pepsinogen is modified post-translationally to form active pepsin.

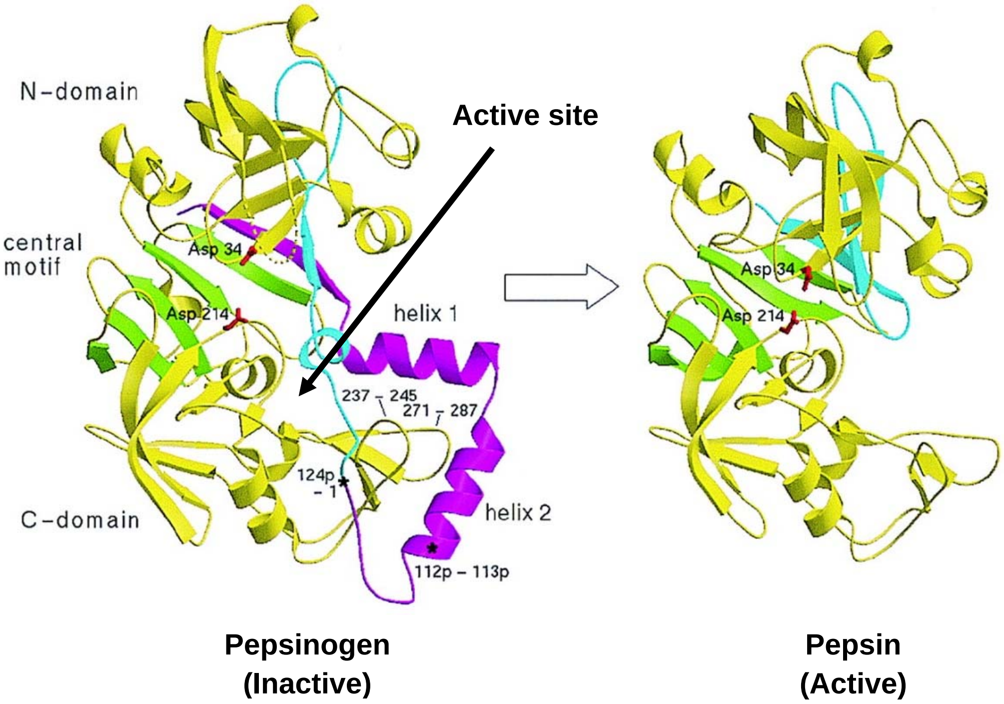


Fig. 2.1

(i) With reference to Fig. 2.1, identify the type of post-translational regulation observed in the enzyme and explain why it results in its activation. [2]

1. Proteolytic cleavage and activation of pepsinogen which removes helix 1 and 2
2. Exposes the active site to allow for substrate to bind and pepsin to catalyse breakdown of proteins

(ii) Suggest why pepsin is synthesised as an inactive precursor pepsinogen. [2]

1. It is a hydrolytic enzyme that catalyses breakdown of protein
2. Hence may degrade essential intracellular protein components (e.g. enzymes, membrane proteins etc.), thus is only activated when it needs to carry out its function

[Total: 17]

3. Bacterial cells can contain one or more plasmids which carry genes that are beneficial to its survival. One such plasmid contains *arg*, *his*, *leu* and *cys* genes that code for the biosynthesis of four essential amino acids; arginine, histidine, leucine and cysteine respectively.

A bacterial strain, strain **A**, with the following plasmid genes (*arg*+, *his*+, *leu*+ and *cys-)* was grown in nutrient media for several generations.

(a) State the amino acid(s), if any, that must be added to the nutrient media for strain **A** to grow. [1]
Cysteine

Strain **A** was then mixed with another bacterial strain, strain **B**, with the following plasmid genes (*arg*-, *his*-, *leu*- and *cys+)* and allowed to grow together at various time intervals before strain **B** was isolated and grown on nutrient media containing a variety of amino acids.

Table 3.1 shows whether colonies of strain **B** were observed on the various media at different time intervals (+ indicates the presence of an amino acid in the medium while - indicates its absence).

Time of incubation/ minutes	Supplementation of amino acids in medium				Presence of colonies
	Arg	His	Leu	Cys	
10	–	+	+	-	No
	+	-	+	-	Yes
	-	-	+	-	No
20	+	-	-	-	Yes
	-	-	-	-	No
25	-	-	-	-	Yes

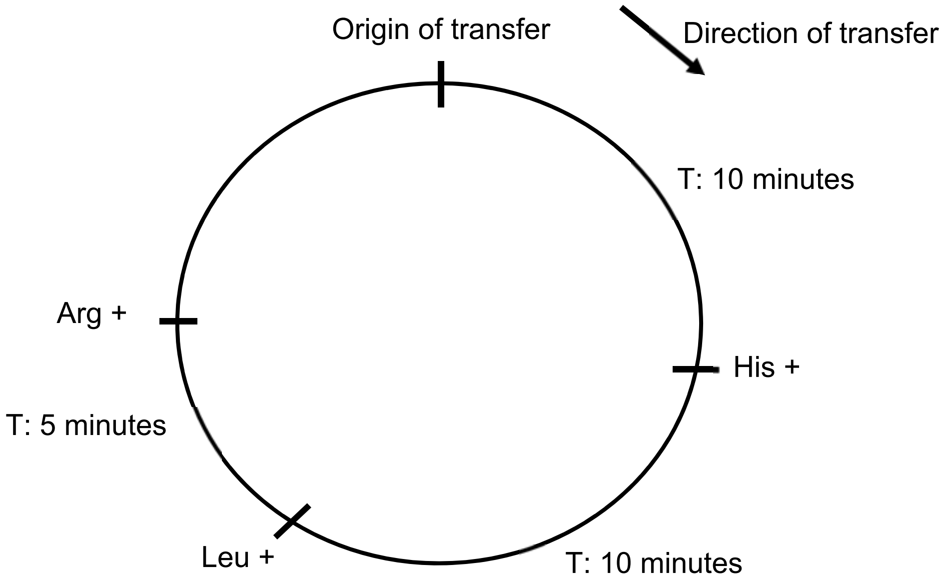
Table 3.1

(b) Describe the process that has occurred between strains **A** and **B** when they were incubated together in the nutrient media. [4]

1. Conjugation
2. Sex pilus from strain A contacts surface of strain B and draws the two cells close
3. Allows formation of cytoplasmic bridge/ conjugation tube between the 2 cells
4. A single strand of the plasmid from strain A was transferred to strain B and used as a template for the synthesis of complementary strand.
5. Strain B now contains 2 double- stranded plasmids

Max 4

(c) Using information from Table 3.1, indicate the order of genes (*arg*+, *his*+, *leu*+) found on the plasmid in strain **A**. You should also indicate in your diagram the origin and direction of transfer. [3]



1. origin and direction of transfer
2. Order of genes
3. Relative distance between genes

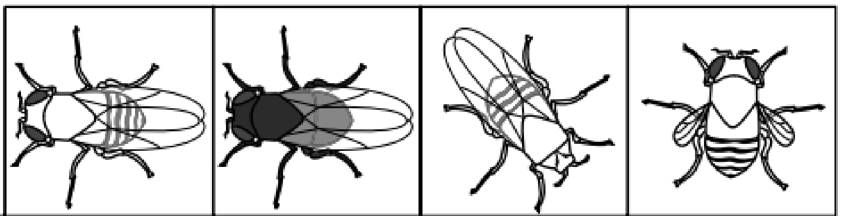
(d) Using information from Table 3.1, explain your answer in **(c).** [3]

1. Genes that are located nearer to the origin of transfer will be transferred to recipient bacteria strain B first and thus would be expressed first
2. As strain B was able to grow in a media that was not supplied with <u>histidine from 10 minutes, histidine and leucine at 20 minutes and none of the three amino acids at 25 minutes/ supplied with only leucine and arginine at 10 minutes, only arginine at 20 minutes and no amino acids at 25 minutes</u>, this indicates that the order of transfer is his+, leu+ and arg+
3. Length of time proportional to the distance between the genes

[Total: 11]

4. The fruit fly, Drosophila melanogaster, has eyes, a striped abdomen and wings longer than its abdomen. This is called a 'wild-type' fly.

Mutation has resulted in many variations of these features. Table 4.1 shows diagrams of a wild-type fly and three other flies, each of which shows one recessive mutation.



eyes	present	present	absent	present
abdomen	striped	black	striped	striped
wing description	long	long	long	short

Table 4.1

(a) Analysis of the mature mRNA formed from the mutant allele for black abdomen showed that one exon was missing although the length of both the mutant and normal alleles for abdomen colour were identical on the chromosome.

Suggest the genetic basis of this mutation and how it leads to a missing exon. [3]

1. Base pair substitution mutation in splice site/ snRNP gene
2. Spliceosome is unable to recognise the mutated splice site sequence and splice out introns properly
3. Leading to excision of the first exon found after mutated splice site

(b) Using appropriate symbols, illustrate a cross between a fly without eyes and long wings and one with eyes and short wings that would result in **four** different phenotypes being observed in the offspring. [5]

Legend [1]
Let E represent the dominant allele for presence of eyes and e the recessive allele for absence of eyes
Let L represent the dominant allele for long wing and l the recessive allele for short wings

<table>
<tr><td>Parental phenotype</td><td>Without eyes, long wings</td><td>X</td><td>With eyes, short wings</td></tr>
<tr><td>Parental genotype: [1]</td><td>eeLl</td><td>X</td><td>Eell</td></tr>
<tr><td>Gametes: [1] *to circle</td><td>eL el</td><td>x</td><td>El el</td></tr>
<tr><td>Punnet square: [1]
*gametes must be circled</td><td colspan="3"><table><tr><td></td><td>eL</td><td>el</td></tr><tr><td>El</td><td>EeLl</td><td>Eell</td></tr></table></td></tr>
</table>

	el	eeLl	Eell
Offspring genotype	EeLl: Eell: eeLl: eell		
Offspring phenotype: [1]	With eyes, long wings: with eyes, short wings: without eyes, long wings: without eyes short wings		
Phenotypic ratio	1:1:1:1		

(c) A cross was carried out between a fly heterozygous for striped abdomen and long wings and a fly with a black abdomen and short wings. The results are shown below in Table 4.2.

offspring	number
striped abdomen long wing	86
black abdomen long wing	87
striped abdomen short wing	81
black abdomen short wing	78
total	332

Table 4.2

A chi-squared test (χ2) was carried out on these data. Complete Table 4.3 and calculate the value of χ2. [3]

Observed Number (O)	Expected Number (E)	(O - E) [1]	$(O - E)^2$	$(O - E)^2$ / E [1]
86	83	3	9	0.11
87	83	4	16	0.19
81	83	-2	4	0.05
78	83	-5	25	0.30

Table 4.3

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

χ2: 0.65 [1]

(d) Table 4.4 shows χ2 values.

degrees of freedom	probability						
	0.50	0.20	0.10	0.05	0.02	0.01	0.001
3	2.37	4.64	6.25	7.82	9.84	11.34	16.27

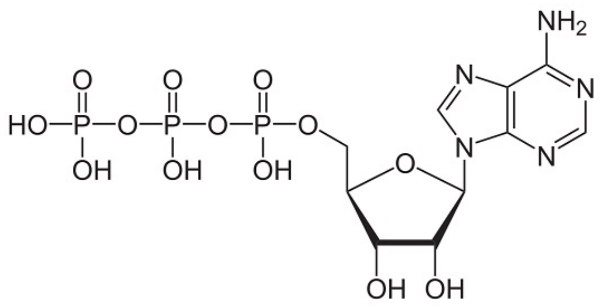
Table 4.4

Using Table 4.4, explain what conclusions can be made about the results of the χ2 test. [2]

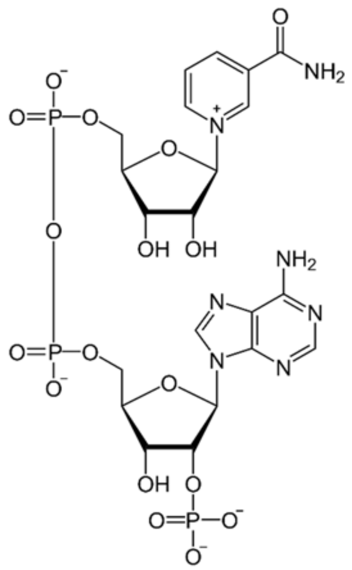
1. Calculated χ2 is less than critical χ2 of 7.82 at probability 0.05.
2. Accept null hypothesis that there is no significant difference between observed and expected results, difference is due to chance.

[Total: 13]

5. ATP and NADP both play important roles in photosynthesis. Fig. 5.1 represents the molecular structures of ATP and NADP.



ATP



NADP

Fig. 5.1

(a) Using Fig. 5.1, compare the structures of ATP and NADP. [4]

1. Both have <u>three</u> phosphate groups
2. Both have an adenine/ purine
3. Both have pentose sugar/ ribose
4. Both have phosphoanhydride bonds

Max 2

5. ATP has one pentose sugar while NADP has two
6. NADP has nicotinamide base which is absent in NADP
7. There are 2 phosphoanhydride bonds present in ATP, but only one present in NADP.

Max 2

(b) Outline the roles of NADP in a cell. [2]

1. Serves as <u>final electron acceptor</u> in non-cyclic photophosphorylation
2. To form <u>reduced NADP/ NADPH</u> which is used in <u>light-independent reaction/ Calvin cycle</u>

(c) State the names of the processes in which ATP is synthesised during photosynthesis and respiration respectively. [2]

1. Photosynthesis: <u>cyclic and noncyclic photophosphorylation</u>
2. Respiration: <u>substrate level phosphorylation and oxidative phosphorylation</u>

(d) ATP serves as a source of energy for several metabolic processes in both photosynthesis and respiration. State two processes in **respiration** that requires ATP as an energy source. [2]

1. Active transport of pyruvate/ acetyl CoA into mitochondrion
2. Phosphorylation of intermediate compounds in glycolysis

(e) The first substrate used in respiration is glucose. In a situation of excess glucose, some of these glucose is stored as fats instead of carbohydrates. Explain why animals prefer to store lipid instead of carbohydrates. [2]

1. Twice the amount of energy produced per gram/ unit mass of lipid/ carbohydrate stored due to higher proportion of C-H to O present
2. Twice the amount of metabolic water per gram/ unit mass of lipid/ carbohydrate oxidised due to higher proportion of C-H to O present
3. Smaller volume stored for the same amount of energy hence aiding in locomotion

4. Ref. to lipids (triglycerides) being good thermal/ heat insulators
5. Ref. to lipids being stored around delicate organs hence acting as cushioning material/ protecting these organs
6. Ref. to lipids being less dense than water, thus providing buoyancy for aquatic animals

Max 1 for points 4 to 6

[Total: 12]

6. Lactose intolerance is a condition in which individuals are unable to digest lactose as the cells in their small intestine are unable to synthesise sufficient quantities of the enzyme lactase. The undigested lactose will then interact with bacteria normally present in the large intestine and cause uncomfortable symptoms such as bloating, diarrhoea and gas production.

Lactose is commonly found in large quantities in dairy products. The distribution of adult population with lactose intolerance is shown in Fig. 6.1. A simplified world map illustrating major continents is shown in Fig. 6.2 for your reference.

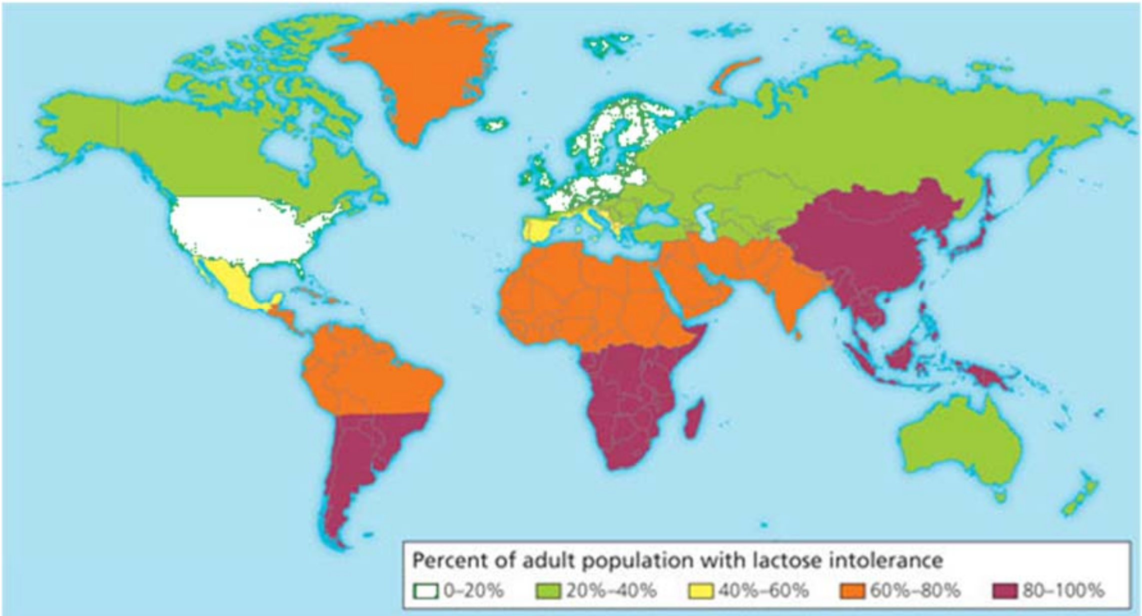


Fig. 6.1

Fig. 6.2

(a) With reference to Figs. 6.1 and 6.2, describe the distribution of lactose intolerance in adults. [2]

1. High lactose intolerance in adult population of 80 to 100% is observed in East Asia, south Africa and southern regions of south America
2. Low lactose intolerance of 0 to 20% is observed in Europe and North America (USA)

(b) Research has shown that most individuals are able to produce large quantities of lactase from birth, although this value can decline as the individual grows older due to environmental factors. The most prominent decline in lactase production is often observed when babies are switched from a predominantly milk-based diet to a solid food diet.

(i) Suggest why there is a prominent decline in lactase production in babies following a switch to a solid food diet. [2]

1. Babies are **originally fed a predominantly milk-based diet**, thus require large quantities of lactase at birth.
2. Following a switch to solid food, amount of dairy consumed declines thus reducing the necessity for large quantities of lactase to be produced.

(ii) Using the information found in **(b),** suggest a reason for your observations made in **(a).** [1]

1. Europe and USA has a diet which consist of much higher quantities of dairy products hence allowing for a smaller reduction is lactase production as the individual grows older.

Level of expression of lactase is regulated by a lactase activator which stimulates expression of lactase in the presence of lactose. Human populations that inhabited the earth millions of years ago were found to express a lactase activator which is only weakly active. At present, most human populations express a more active version of this activator protein.

(c) Using your knowledge of evolution, explain the basis of this observation. [4]

1. Variation exist in the population in terms of the gene coding for the activator protein
2. Selection pressure in the form of diet high in dairy products containing lactose
3. Individuals that express the more active lactase activator protein are better able to produce higher quantities of lactase and hence digest lactose effectively to obtain nutrients are at a selective advantage, while those that cannot are at selective disadvantage in a diet high in dairy
4. Favourable allele for more active lactase activator passed on to offsprings, changing frequency of allele in the population over time

[Total: 9]

7. Antigen presenting cells (APCs) such as macrophages are able to detect specific foreign antigens and present them to relevant adaptive immune cells. Recognition of these antigens require the use of specific receptors on the surface of these APCs.

(a) State the receptor found on macrophages that is used for antigen binding and recognition. [1]

Toll like receptors

Following binding, these antigens are then taken into the cell and processed for presentation to T cells.

(b) Describe how an **extracellular** antigen is subsequently presented to a T cell following binding to the receptor mentioned in **(a)**. [3]

1. Antigen taken into the cell by receptor-mediated endocytosis and processed into short peptide fragments by phagocytosis
2. A peptide fragment binds to MHC Class II complex produced by rough endoplasmic reticulum of APC
3. Peptide:class II MHC complex transported to cell surface membrane for antigen presentation to CD4/ helper T cell.

Research has shown that a cell signalling pathway is triggered following binding of an antigen to its receptor. One of the more well understood signalling pathways involve a series of kinases, which are enzymes that catalyse the phosphorylation of its substrate. A common signalling pathway in APCs is shown in Fig. 7.1.

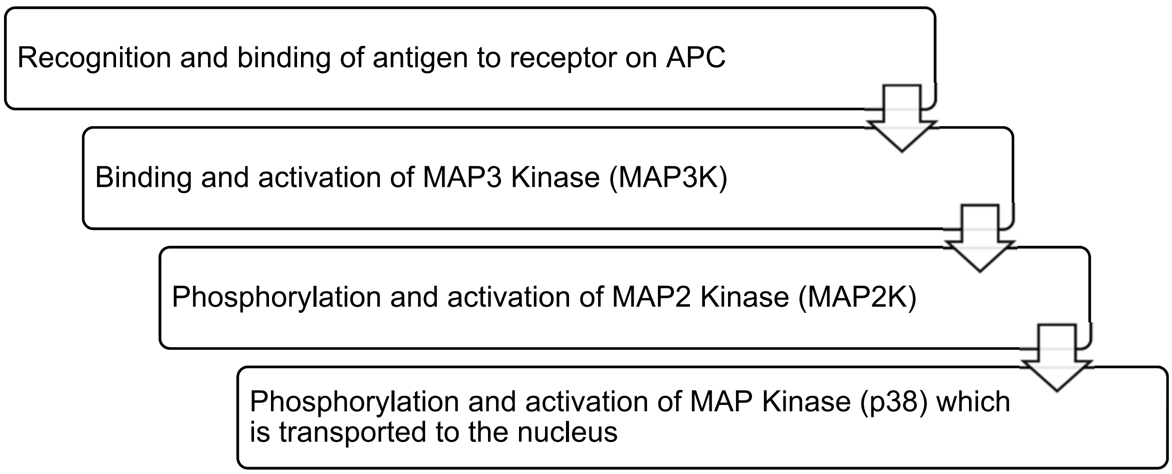


Fig. 7.1

Activation of the signalling pathway shown in Fig. 7.1 leads to a large cellular response consisting of both cytokine production and also antigen presentation.

(c) Using your knowledge of signalling pathways and the information from Fig. 7.1, explain this observation. [4]

1. Activation of protein kinases (MAP3K and MAP2K)
2. triggers a phosphorylation cascade
3. that amplifies the signal

4. More molecules are phosphorylated and activated in the subsequent step compared to the preceding step
5. p38 then triggers expression/ transcription of genes that code for cytokines and MHC molecules
   *Mark once for reference to MAP3K/2K

Max 4

[Total: 8]

8. Dengue and tuberculosis (TB) are two prominent infectious diseases of concern in many countries.

(a) Describe how TB is transmitted from an infected person to an uninfected person. [2]

1. Airborne disease transmitted via aerosol/ droplets produced by infected person when they cough/ sneeze/ speak
2. Subsequent inhalation of respiratory secretions that contain TB bacterium by uninfected person

Fig. 8.1 shows the distribution of dengue.

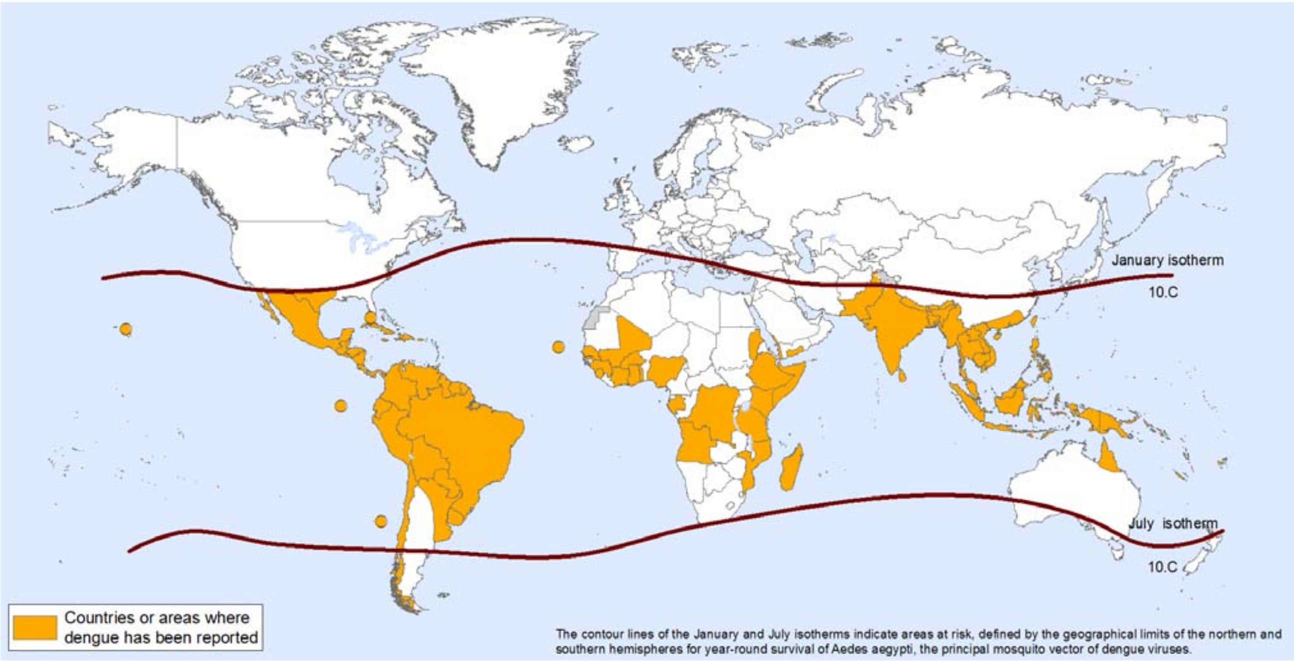


Fig. 8.1

(b) Unlike dengue, TB is found across the entire world. Explain why dengue shows the distribution shown in Fig. 8.1 whereas TB is found worldwide. [3]

1. Dengue virus needs to reproduce within mosquito vector as part of its reproductive life cycle
2. *Aedes* mosquito vector lives predominantly in tropics/ hot areas with rainfall
3. TB only requires a human host where internal body temperature is constant

Vaccinations are used to control infectious diseases. They were used as part of the programme to eradicate smallpox and as part of the continuing programmes against diseases such as polio and measles.

Smallpox was eradicated from the world in the 1970s. Polio is likely to be the next infectious disease to be eradicated.

(c) Explain how vaccination provides immunity as an important part of programmes to control and eradicate infectious diseases. [5]

1. Vaccination involve introduction of antigens in a safe way to induce the adaptive immune response
2. A portion of these cells form memory cells, giving long term immunity/ immunological memory
3. Leads to a stronger and faster secondary adaptive immune response to the pathogens upon second exposure
4. Herd immunity is acquired
5. as immunised individual will not spread the disease to other unimmunised individual

Despite being a disease that has persisted for hundreds of years, there is currently no vaccine approved for the treatment of TB.

(d) Using your knowledge of the pathogenicity of *M. tuberculosis*, suggest why it is difficult to develop an effective vaccine for TB. [1]

1. *M. tuberculosis* bacteria spends most of its time hidden within macrophages as it replicates hence evading immune cells and antibodies.
2. Bacteria can remain dormant in tubercle for years and remain undetected by immune cells.
3. Mutation in *M. tuberculosis* changes epitopes recognised by antibodies thus requiring repeated/ new vaccination.
4. AVP

Apart from the use of vaccination, other measures of controlling the spread of dengue involves the release of sterile male mosquitoes into areas with high dengue incidence. While this measure has successfully reduced mosquito populations dramatically, environmentalist are concerned about their potential detrimental ecological effects.

(e) Suggest one possible ecological concern that may arise from the use of sterile male mosquitoes. [1]

1. Decline is populations that are directly dependent on *Aedes* mosquitoes for food
2. Loss of pollinators for plants that rely on mosquitos to pollinate them > loss in biodiversity
3. AVP

[Total: 12]